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Comprehensive Question Bank: Unit 6 — Emerging Technologies

Section A: Multiple-Choice Questions (MCQs)

Topic 6.1: Definition and Overview of Emerging Technologies

1. Emerging technologies are best defined as:
 - A) Tools that have existed for over a century and no longer change
 - **B) New tools, systems, and methods currently being developed or only recently used, with the potential to change how we live and work**
 - C) Only technologies related to the internet
 - D) Technologies used exclusively by large corporations
2. Which emerging technology refers to machines and software that can learn and perform tasks similar to human beings, such as recognizing faces or understanding speech?
 - A) Cloud Computing
 - **B) Artificial Intelligence (AI)**
 - C) Blockchain
 - D) 5G Technology
3. Which technology allows people to store and access data and applications over the internet instead of on a local computer?
 - A) Internet of Things (IoT)
 - **B) Cloud Computing**
 - C) Blockchain
 - D) Augmented Reality
4. Which technology is best known as the foundation behind cryptocurrencies like Bitcoin, but is also used in supply chains and contracts?
 - A) Cloud Computing
 - B) Artificial Intelligence
 - **C) Blockchain**
 - D) Quantum Computing
5. Which technology connects everyday objects, like refrigerators and cars, to the internet so they can send and receive data?
 - A) 5G Technology

- **B) Internet of Things (IoT)**
 - C) Quantum Computing
 - D) Biotechnology
6. What is the key difference between Augmented Reality (AR) and Virtual Reality (VR)?
- A) AR and VR are identical technologies with different names
 - **B) AR adds digital elements to the real world, while VR creates a completely virtual environment**
 - C) VR only works on smartphones, while AR requires special glasses
 - D) AR is used only in gaming, while VR is used only in education
7. What is the main benefit of 5G technology?
- A) It replaces the need for the internet entirely
 - **B) It offers much faster internet speeds and more reliable connections**
 - C) It eliminates the need for wireless devices
 - D) It only benefits smart thermostats
8. What is a "qubit" in quantum computing?
- A) A regular bit that can only be 0 or 1
 - **B) A tiny building block of quantum computers that can represent 0 and 1 at the same time**
 - C) A unit used to measure internet speed
 - D) A type of cloud storage unit
9. Biotechnology primarily involves:
- A) Building faster computer processors
 - **B) Using living organisms, like bacteria and plants, to create new products or solve problems**
 - C) Creating virtual reality environments
 - D) Managing peer-to-peer networks
-

Topic 6.2: Cloud Computing (Overview)

1. Cloud computing is best described as:
- A) A method of permanently storing data only on local hard drives
 - **B) A model allowing easy and convenient access to computing resources like servers, storage, and applications over the internet**
 - C) A type of physical computer hardware
 - D) A programming language used for web development
2. Which analogy best describes cloud computing, according to this unit?
- A) Digging your own well for water
 - **B) Renting a supercomputer that you can use whenever you need it, from anywhere in**

the world

- C) Building your own private road network
 - D) Writing a letter and mailing it physically
-

Topic 6.2.1: Basic Concepts of Cloud Computing

1. Virtualization is best described as:
 - A) A technology that deletes unused files automatically
 - **B) A technology that allows a single physical machine to run multiple virtual machines**
 - C) A method of encrypting cloud data
 - D) A type of internet connection
2. What software layer manages and creates virtual machines on top of physical hardware?
 - A) Operating system only
 - **B) Hypervisor**
 - C) Web browser
 - D) Firewall
3. Each virtual machine created through virtualization can:
 - A) Only run one shared operating system with all other VMs
 - **B) Run its own operating system and applications as if it were an independent machine**
 - C) Never access the internet
 - D) Only be used by the cloud provider, not customers
4. Scalability in cloud computing refers to:
 - A) Automatic, real-time adjustment of resources with no human involvement
 - **B) The ability to add more resources when needed, such as during a sales event**
 - C) A method of encrypting data before storage
 - D) The process of virtualizing a single server
5. Elasticity in cloud computing refers to:
 - A) Manually adding servers weeks in advance of an anticipated event
 - **B) A cloud system's ability to automatically scale resources up or down based on current demand**
 - C) The physical flexibility of server hardware
 - D) A fixed, unchangeable allocation of resources
6. Which analogy from this unit best describes elasticity?
 - A) Buying a bigger jacket in advance for winter
 - **B) An elastic waistband that stretches automatically when you eat a big meal**
 - C) Digging a well before drinking water
 - D) Building a private swimming pool

7. On-demand access in cloud computing means:
- A) Resources are only available after a lengthy setup process
 - **B) You can use computing resources whenever you need them, without a long setup process**
 - C) Resources can only be accessed once per day
 - D) Only administrators can request new resources
8. Which analogy from this unit is used to describe on-demand access?
- A) An elastic waistband
 - **B) Turning on a tap to get water instantly, instead of digging a well first**
 - C) Renting an apartment
 - D) A vending machine
-

Topic 6.2.2: Types of Cloud Services

1. Which of the following is NOT one of the three main types of cloud services discussed in this unit?
- A) Infrastructure as a Service (IaaS)
 - B) Platform as a Service (PaaS)
 - C) Software as a Service (SaaS)
 - **D) Data as a Service (DaaS)**
2. IaaS (Infrastructure as a Service) offers:
- A) Fully finished software applications ready to use immediately
 - **B) Basic computing infrastructure such as servers, storage, and networking on a pay-as-you-go basis**
 - C) Only development tools with no infrastructure
 - D) Physical hardware ownership to the customer
3. In the IaaS model, users have control over:
- A) The underlying physical infrastructure only
 - **B) The operating systems, applications, and storage, but not the underlying physical infrastructure**
 - C) Nothing at all — everything is managed by the provider
 - D) Only the internet connection speed
4. Which of the following is an example of an IaaS provider mentioned in this unit?
- A) Google Workspace
 - **B) Amazon Web Services (AWS)**
 - C) Microsoft Office 365
 - D) Salesforce

5. PaaS (Platform as a Service) provides:
- A) Only physical servers with no development tools
 - **B) A complete development and deployment environment, including infrastructure, middleware, development tools, and management services**
 - C) Only end-user ready software with no coding involved
 - D) Just a web browser for accessing files
6. Which of the following is an example of a PaaS provider mentioned in this unit?
- A) AWS EC2
 - **B) Google App Engine**
 - C) Google Drive
 - D) Dropbox
7. SaaS (Software as a Service) is best described as:
- A) Renting raw servers and storage
 - B) A development environment for coding applications
 - **C) Access to software applications that are hosted and managed by the service provider, used directly over the internet**
 - D) A type of cloud deployment model
8. Which of the following is an example of a SaaS application mentioned in this unit?
- A) AWS EC2
 - B) Google App Engine
 - **C) Google Workspace (Gmail, Google Docs, Google Drive)**
 - D) Google Compute Engine
9. Using the "Pizza as a Service" analogy, which cloud service model is comparable to buying raw ingredients and baking a pizza yourself at home?
- **A) IaaS**
 - B) PaaS
 - C) SaaS
 - D) None of these
10. Using the "Pizza as a Service" analogy, which cloud service model is comparable to going to a restaurant where the entire meal is cooked and served ready to eat?
- IaaS
 - PaaS
 - C) SaaS**
 - D) None of these
11. Which cloud service model requires the LEAST management effort from the end-user?
- IaaS
 - PaaS

- C) SaaS**
 - D) All require equal effort
12. Which cloud service model gives developers the MOST control over the operating system and infrastructure setup?
- A) IaaS**
 - PaaS
 - SaaS
 - D) None of these
-

Topic 6.2.3: Cloud Deployment Models

1. A public cloud is best described as:
 - A) A cloud environment used exclusively by one organization
 - **B) A cloud service offered over the internet and shared among multiple organizations, managed by a third-party provider**
 - C) A cloud that combines public and private features
 - D) A strategy using multiple cloud providers simultaneously
2. Which of the following is an example of a public cloud provider mentioned in this unit?
 - A) A bank's internal private servers
 - **B) Amazon Web Services (AWS)**
 - C) A hospital's on-premises data center
 - D) None of these
3. A private cloud is best described as:
 - A) A cloud service shared among multiple organizations
 - **B) A cloud environment used exclusively by one organization, whether hosted on-premises or by a third party**
 - C) A combination of public and private cloud features
 - D) A cloud strategy using multiple providers at once
4. Which scenario best illustrates the use of a private cloud, according to this unit?
 - A) A small blog hosted on a shared public server
 - **B) A large bank handling sensitive customer data securely within its own or a dedicated third-party data center**
 - C) A retail company using three different cloud providers
 - D) A student using free cloud storage for school photos
5. A hybrid cloud is best described as:
 - A) A cloud used by only one organization
 - **B) A model that combines public and private clouds, allowing data and applications to**

be shared between them

- C) A cloud service with no internet connectivity
 - D) A cloud strategy that avoids using any private infrastructure
6. Which scenario best illustrates hybrid cloud usage, according to this unit?
- A) A company using only AWS for everything
 -) A company using a public cloud for everyday operations while keeping sensitive data in a private cloud, shifting load between them during busy - iods**
 - C) A company using zero cloud services at all
 - D) A company that only uses on-premises hardware
7. A multi-cloud model is best described as:
- A) Using only one cloud provider for all needs
 - **B) A strategy where an organization uses services from multiple cloud providers simultaneously to meet different business or technical needs**
 - C) Combining public and private cloud into a single environment
 - D) A cloud used only for backup purposes
8. In the multi-cloud example from this unit, which provider does the global retail company use for advanced data analytics and machine learning services?
- A) Amazon Web Services (AWS)
 - B) Microsoft Azure
 - **C) Google Cloud Platform (GCP)**
 - D) Salesforce
9. According to the comparison in this unit, which deployment model is described as cost-effective but less secure?
- **A) Public Cloud**
 - B) Private Cloud
 - C) Hybrid Cloud
 - D) Multi-Cloud
10. According to the comparison in this unit, which deployment model is described as more secure but expensive?
- Public Cloud
 - B) Private Cloud**
 - Multi-Cloud
 - D) None of these
11. According to the comparison in this unit, which deployment model provides resilience by avoiding dependence on a single provider?
- Public Cloud
 - Private Cloud

- Hybrid Cloud
 - **D) Multi-Cloud**
12. Which deployment model is described as offering flexibility by combining two different environments?
- Public Cloud
 - Private Cloud
 - C) Hybrid Cloud**
 - D) Multi-Cloud
13. For a hospital needing to securely store patient records while also using AI tools on flexible, powerful public infrastructure, which deployment model is generally most appropriate?
- Public Cloud only
 - Private Cloud only
 - C) Hybrid Cloud**
 - D) None of these would work
-

Topic 6.3: Applications and Implications of Cloud Computing

1. Cloud storage allows users to:
 - A) Only store data on local devices
 - **B) Save data on remote servers rather than on local devices, making it easier to access and share**
 - C) Permanently delete data after one use
 - D) Avoid using the internet entirely
2. Which of the following is an example of a cloud storage service mentioned in this unit?
 - A) AWS Lambda
 - **B) Google Drive**
 - C) Ethereum
 - D) Cloudflare (as a CDN specifically)
3. What is the primary role of a Content Delivery Network (CDN), such as Cloudflare?
 - A) To store cryptocurrency transactions
 - **B) To deliver website content quickly by caching it on servers close to end-users**
 - C) To virtualize physical servers
 - D) To manage blockchain consensus
4. Which cloud-based platforms are mentioned in this unit as tools for building, training, and deploying machine learning models?
 - A) Google Drive and Dropbox

- **B) Google Cloud AI and AWS SageMaker**
 - C) Ethereum and Bitcoin
 - D) AWS Lambda and Cloudflare
5. Which of the following is a significant concern in cloud computing due to storing sensitive data on remote servers?
- A) Increased hardware costs only
 - **B) Data security, including risks like data breaches and unauthorized access**
 - C) Decreased internet speed
 - D) Loss of internet connectivity
6. Which of the following is recommended in this unit as a security measure to mitigate cloud data risks?
- A) Avoiding all cloud services entirely
 - **B) Using encryption, strong authentication methods, and regularly reviewing security policies**
 - C) Sharing passwords with all team members
 - D) Disabling all monitoring tools
7. Effective resource management in cloud computing helps to:
- A) Increase unnecessary costs deliberately
 - **B) Control costs and ensure efficient use of cloud resources**
 - C) Prevent any scaling of resources
 - D) Eliminate the need for monitoring
8. Regarding cost considerations in cloud computing, users typically:
- A) Pay a single fixed fee regardless of usage
 - **B) Pay for what they use, meaning costs can add up quickly if not monitored**
 - C) Never need to review their spending
 - D) Receive cloud services for free permanently
9. Compliance and regulatory issues in cloud computing refer to:
- A) The physical location of a company's headquarters only
 - **B) Ensuring cloud service usage complies with legal and regulatory requirements, which vary by region and industry**
 - C) A type of cloud deployment model
 - D) The technical process of virtualization
-

Topic 6.4: Introduction to Blockchain Technology

1. Blockchain technology is best described as:
- A) A single centralized database controlled by one company
 - **B) A revolutionary concept enabling secure and transparent transactions through a**

distributed ledger system

- C) A type of cloud deployment model
 - D) A programming language for web development
2. According to the "digital notebook" analogy in this unit, what happens when someone makes a change, like paying back money, in a blockchain system?
- A) Only one person's notebook is updated
 - **B) It gets recorded in all the notebooks (copies of the ledger) at the same time**
 - C) The change is kept secret from other participants
 - D) The notebook is destroyed and a new one is created
3. Which special feature of blockchain ensures that everyone in the group can see what's written, making it hard for anyone to cheat unnoticed?
- A) Security
 - **B) Transparency**
 - C) Decentralization
 - D) Consensus
4. Which special feature of blockchain makes it almost impossible to erase or change information once written, due to cryptography?
- A) Transparency
 - **B) Security**
 - C) Decentralization
 - D) Virtualization
5. Which special feature of blockchain means there is no single person or computer in charge, and changes are only made when the majority agree?
- A) Transparency
 - B) Security
 - **C) Decentralization**
 - D) Elasticity
-

Topic 6.4.1: Fundamentals of Blockchain (Core Principles, Components, P2P Networks)

1. Which core principle of blockchain states that it is maintained by a network of computers (nodes) rather than a central authority?
- A) Immutability
 - **B) Decentralization**
 - C) Consensus Mechanism
 - D) Virtualization

2. Which core principle of blockchain ensures that once a block is added, it cannot be altered or deleted?
- A) Decentralization
 - **B) Immutability**
 - C) Consensus Mechanism
 - D) Scalability
3. What do consensus mechanisms in a blockchain network ensure?
- A) Only one node validates all transactions
 - **B) All nodes in the network reach a unanimous decision before adding a new block to the chain**
 - C) Transactions are hidden from most participants
 - D) Blocks can be deleted upon request
4. In blockchain terminology, a "node" refers to:
- A) A unit of cryptocurrency
 - **B) A computer that participates in the blockchain network, maintaining a copy of the blockchain and helping validate transactions**
 - C) A type of smart contract
 - D) A cloud deployment model
5. The "ledger" in a blockchain system refers to:
- A) A single private company database
 - **B) The shared digital record of all transactions that have occurred on the blockchain**
 - C) A type of cryptocurrency wallet
 - D) A cloud storage service
6. A "block" in a blockchain is best described as:
- A) A single transaction only
 - **B) A collection of transactions bundled together, containing a unique hash, a reference to the previous block's hash, and a list of transactions**
 - C) A type of virtual machine
 - D) A cloud deployment model
7. What does the "parent hash" inside a block refer to?
- A) A random number with no significance
 - **B) A reference to the previous block's unique hash, linking blocks together in a chain**
 - C) The identity of the block's creator
 - D) The total number of transactions in the network
8. The "blockchain protocol" is best described as:
- A) A single company's internal policy document
 - **B) The rules and procedures defining how transactions are validated, how blocks are**

added, and how consensus is achieved

- C) A type of cryptocurrency
- D) A cloud service model

9. A peer-to-peer (P2P) network is best described as:

- A) A system relying entirely on one central server for all communication
- **B) A system where computers (nodes) communicate and share resources directly with each other, without relying on a central server**
- C) A type of cloud deployment model
- D) A method of encrypting cryptocurrency wallets

10. In a peer-to-peer network, each node can act as:

- Only a client, never a server
 - Only a server, never a client
 - C) Both a client and a server**
 - D) Neither a client nor a server

11. In blockchain, what is the significance of a cryptographic hash changing completely when even one small detail in a block is altered?

- It has no significance and is purely coincidental
- B) It immediately reveals tampering, since every following block's reference to the previous hash would no longer match**
 - It automatically deletes the tampered block
 - D) It only affects the node that made the change

Topic 6.4.2: Use Cases of Blockchain Technology

1. Which of the following is described in this unit as blockchain's best-known use case?

- A) Web hosting
- **B) Cryptocurrencies, such as Bitcoin and Ethereum**
- C) Cloud storage
- D) Virtual machines

2. How can blockchain be used in supply chain management, according to this unit?

- A) To increase the cost of shipping goods
- **B) To track and verify the movement of goods, helping prevent fraud and ensure product authenticity**
- C) To replace all physical warehouses
- D) To eliminate the need for manufacturers

3. How can blockchain be used in healthcare, according to this unit?
 - A) To make patient records publicly visible to everyone
 - **B) To securely store patient records and ensure only authorized individuals access sensitive information**
 - C) To replace doctors with automated systems
 - D) To eliminate hospitals entirely
 4. How can blockchain be used in voting systems, according to this unit?
 - A) To allow unlimited votes per person
 - **B) To create secure and transparent voting systems, ensuring votes are accurately recorded and reducing election fraud**
 - C) To make voting records completely private and unverifiable
 - D) To replace the need for elections altogether
-

Topic 6.4.3: Cryptocurrencies and Smart Contracts

1. Cryptocurrencies are best described as:
 - A) Physical coins issued only by government banks
 - **B) Digital currencies that work without traditional banks, allowing direct transactions between people worldwide**
 - C) A type of cloud deployment model
 - D) A programming language for smart contracts
2. What advantage do cryptocurrencies offer compared to traditional bank-issued money, according to this unit?
 - A) They require more intermediaries to process transactions
 - **B) They offer a secure and decentralized way to exchange money, with transactions that cannot be changed**
 - C) They can only be used within a single country
 - D) They require a central authority to approve every transaction
3. A smart contract is best described as:
 - A) A paper-based legal agreement signed by a lawyer
 - **B) A digital agreement, written in code, that automatically carries out its terms when specific conditions are met**
 - C) A type of cryptocurrency wallet
 - D) A method of virtualizing servers
4. Which platform is mentioned in this unit as enabling developers to create decentralized applications (DApps) using smart contracts?
 - A) AWS Lambda

- **B) Ethereum**
 - C) Google App Engine
 - D) Cloudflare
5. Which of the following is mentioned as a real challenge associated with smart contracts?
- A) They cannot be used for any financial transactions
 - **B) The need for error-free code and legal systems to resolve related disputes**
 - C) They require a bank to approve every execution
 - D) They cannot run on blockchain technology
-

Topic 6.5: Applications and Implications of Blockchain

1. How does blockchain help track the origin of products through a supply chain, according to this unit?
 - A) By hiding transaction details from all parties- **B) By recording every transaction on a decentralized ledger, making each step from raw - material supplier to final customer traceable and utable**
 - C) By allowing only the manufacturer to see product history
 - D) By deleting old transaction records regularly
2. According to the "Do You Know?" note in this unit, how do some artists use blockchain?
 - A) To physically print their artwork
 - **B) To sell digital art, where each piece has a unique digital signature proving authenticity and originality**
 - C) To store their artwork exclusively on private clouds
 - D) To avoid paying for cloud storage
3. How does blockchain benefit financial services, according to this unit?
 - A) By making international money transfers slower and more expensive
 - **B) By making transactions faster and safer, especially for sending money abroad**
 - C) By eliminating the need for any identity verification
 - D) By requiring more intermediary banks for each transaction
4. In the "Sending a Secure Letter" analogy, what does "sealing the letter in a special envelope only your friend can open" represent in blockchain?
 - A) Decentralization
 - **B) Encryption**
 - C) Consensus mechanism
 - D) A blockchain protocol
5. In the "Sending a Secure Letter" analogy, what does "signing the envelope with your unique signature" represent in blockchain?
 - A) Decentralization

- B) Encryption
 - **C) Digital signature**
 - D) Virtualization
6. In the "Sending a Secure Letter" analogy, what does "sending copies of the letter through different trusted delivery services to multiple locations" represent in blockchain?
- A) Encryption
 - B) Digital signature
 - **C) Decentralization**
 - D) Consensus mechanism
7. According to the "Do You Know?" note in this unit, which companies use their powerful computers to help run blockchain networks?
- A) Only small startups
 - **B) Big companies like Amazon and Microsoft**
 - C) Only government agencies
 - D) No companies — blockchain requires no computing power
-

Topic 6.6: Future Trends and Innovations

1. Edge computing is best described as:
- A) Processing data only in a centralized, distant data center
 - **B) Bringing processing power closer to data sources, reducing latency and improving efficiency**
 - C) A type of cryptocurrency
 - D) A cloud deployment model combining public and private clouds
2. What does "latency" refer to, in the context of edge computing?
- A) The total storage capacity of a server
 - **B) The delay between an action and a system's response**
 - C) The cost of cloud computing resources
 - D) The number of nodes in a blockchain network
3. According to this unit, how does edge computing benefit autonomous vehicles?
- A) By sending all sensor data to a distant data center before responding
 - **B) By allowing data from sensors and cameras to be processed locally in the vehicle, enabling quick responses to changing road conditions**
 - C) By eliminating the need for sensors altogether
 - D) By increasing the time needed to respond to road conditions
4. Which of the following applications is mentioned in this unit as especially benefiting from edge computing's low latency?
- A) Long-term data archiving

- **B) Smart cities, healthcare monitoring, and industrial automation**
 - C) Static website hosting only
 - D) Email storage
5. Serverless architectures allow developers to:
- A) Run applications with no servers involved at any level, anywhere
 - **B) Build and deploy applications without personally managing servers, while cloud providers automatically allocate resources**
 - C) Avoid the internet entirely
 - D) Only use private cloud infrastructure
6. In a serverless model, how are developers typically billed?
- A) A single fixed annual fee regardless of usage
 - **B) Based on the actual usage of computing resources**
 - C) Based on the number of employees at their company
 - D) They are never billed under any circumstances
7. Which AWS service is mentioned in this unit as an example of serverless computing?
- A) AWS SageMaker
 - **B) AWS Lambda**
 - C) Amazon S3
 - D) AWS EC2
8. What is a common but important clarification about the term "serverless," according to this unit's teaching philosophy?
- A) It means there are absolutely no physical servers anywhere
 - **B) It does not mean there are no physical servers — it means the developer doesn't have to manage them directly**
 - C) It means the application runs entirely offline
 - D) It means the cloud provider charges nothing at all
-

Section B: Short Answer Questions (SQs)

Topic 6.1: Definition and Overview of Emerging Technologies

1. Define "emerging technologies" in your own words.
2. List four emerging technologies discussed in this unit, along with one real-world example of each.
3. Differentiate between Augmented Reality (AR) and Virtual Reality (VR).
4. Explain briefly what a "qubit" is and how it differs from a regular bit.

5. Identify which emerging technology is most closely associated with Bitcoin, and briefly explain why.

Topic 6.2 & 6.2.1: Cloud Computing Basic Concepts

1. Define cloud computing in your own words.
2. Define virtualization and explain the role of a hypervisor.
3. Differentiate between scalability and elasticity in cloud computing.
4. Explain briefly what "on-demand access" means, using the analogy given in this unit.
5. Predict what might happen to a website without scalability or elasticity during a sudden traffic surge.

Topic 6.2.2: Types of Cloud Services

1. Define IaaS, PaaS, and SaaS in your own words.
2. Differentiate between IaaS and PaaS in terms of what the user manages versus what the provider manages.
3. Identify one real-world example each of an IaaS, PaaS, and SaaS provider.
4. Explain briefly why SaaS requires the least technical management from the end-user.
5. Using the "Pizza as a Service" analogy, briefly explain which service model matches "ordering a pizza kit with the crust and oven already provided."

Topic 6.2.3 & 6.2.4: Cloud Deployment Models

1. Define public cloud and private cloud, and briefly state one difference between them.
2. Differentiate between a hybrid cloud and a multi-cloud strategy.
3. Explain briefly why a bank might choose a private cloud over a public cloud.
4. State one advantage and one disadvantage of using a public cloud.
5. Identify which deployment model would best suit an organization wanting resilience by avoiding dependence on a single provider.

Topic 6.3: Applications and Implications of Cloud Computing

1. Define cloud storage and give one real-world example.
2. Explain briefly the role of a Content Delivery Network (CDN) in web hosting.
3. Identify two cloud-based platforms mentioned in this unit for machine learning and AI development.
4. Explain briefly why data security is a significant concern in cloud computing.
5. List two security measures users can take to protect their data in the cloud.

6. Explain briefly why cost management is important, even though cloud computing can be cost-effective.
7. Define what compliance and regulatory issues mean in the context of cloud computing.

Topic 6.4 & 6.4.1: Blockchain Fundamentals

1. Define blockchain technology using the "digital notebook" analogy from this unit.
2. Differentiate between transparency, security, and decentralization as blockchain's three special features.
3. Define the term "immutability" as it relates to blockchain, and briefly explain why it is important.
4. Differentiate between a node, a block, and a transaction in blockchain terminology.
5. Explain briefly the role of the "parent hash" in linking blocks together.
6. Define a peer-to-peer (P2P) network, and briefly explain how it differs from a traditional client-server model.
7. Analyze the role of Peer-to-Peer Networks in Blockchain. How do they function, and why are they essential?

Topic 6.4.2 & 6.4.3: Blockchain Use Cases, Cryptocurrencies, and Smart Contracts

1. List two use cases of blockchain technology beyond cryptocurrency.
2. Explain briefly how blockchain can improve supply chain management.
3. Define cryptocurrency and briefly explain its role in the digital economy.
4. Define a smart contract in your own words.
5. Explain briefly one real challenge associated with using smart contracts.
6. Identify one platform mentioned in this unit that supports smart contracts and decentralized applications.

Topic 6.5: Applications and Implications of Blockchain

1. Explain briefly how blockchain helps track the origin of products in a supply chain.
2. Explain briefly how blockchain benefits financial services, especially international money transfers.
3. Using the "Sending a Secure Letter" analogy, differentiate between encryption and a digital signature.
4. Explain briefly what decentralization means in the "Sending a Secure Letter" analogy from this unit.
5. Describe the concept of immutability in blockchain. Why is it a critical feature?

Topic 6.6: Future Trends and Innovations

1. Define edge computing in your own words.
 2. Explain briefly the meaning of "latency," and why it matters for real-time applications.
 3. What is edge computing and how does it benefit data processing?
 4. How does edge computing improve the efficiency of autonomous vehicles?
 5. Describe the concept of serverless architectures.
 6. What advantages do serverless architectures offer to developers?
 7. Explain briefly why the term "serverless" can be misleading, according to this unit.
 8. Differentiate between Elasticity and On-Demand access in cloud computing.
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Section C: Long / Extensive Questions (LQs)

Topic 6.1: Definition and Overview of Emerging Technologies

1. Discuss in detail the landscape of emerging technologies introduced in this unit.
 - a) Define and briefly describe each of the eight emerging technologies discussed (AI, Cloud Computing, Blockchain, IoT, AR/VR, 5G, Quantum Computing, Biotechnology).
 - b) Analyze how at least three of these technologies might work together in a single real-world system, such as a smart city.
 - c) Evaluate which of these emerging technologies you believe will have the greatest impact on daily life over the next decade, and justify your reasoning.

Topic 6.2, 6.2.1 & 6.2.2: Cloud Computing Concepts and Service Models

1. Classify the various types of cloud services and compare them, highlighting key distinctions.
 - a) Define IaaS, PaaS, and SaaS, explaining what the user controls versus what the provider controls in each model.
 - b) Construct a detailed comparison table covering control, cost, management responsibility, and real-world examples for all three service models.
 - c) Using the "Pizza as a Service" analogy, justify which service model would be most appropriate for a small startup with limited technical staff, and which would suit a large enterprise with a dedicated IT team.
2. Analyze the foundational concepts of cloud computing — virtualization, scalability, elasticity, and on-demand access — and their combined role in enabling modern cloud services.
 - a) Explain, in detail, how virtualization allows a single physical server to serve multiple independent customers.
 - b) Differentiate between scalability and elasticity using a real-world scenario involving a

sudden traffic surge on an e-commerce website.

c) Evaluate why on-demand access is considered one of cloud computing's most transformative features compared to traditional IT infrastructure purchasing.

Topic 6.2.3 & 6.2.4: Cloud Deployment Models

1. Define cloud deployment models and assess the differences among them.
 - a) Describe public, private, hybrid, and multi-cloud deployment models in detail, including a real-world example of each.
 - b) Construct a comparison table evaluating each deployment model across control, security, and cost.
 - c) Evaluate a real-world scenario — a hospital needing both secure patient record storage and flexible AI-powered image analysis — and justify which deployment model(s) would best serve its needs.
 - scribe "Cloud Deployment Models" with examples.
 - a) Explain the key differences between hybrid cloud and multi-cloud strategies, providing a real-world example of each.
 - b) Discuss why an organization might deliberately choose to use more than one cloud provider (multi-cloud) rather than committing to a single - provider.
- c) Analyze a scenario where a global retail company uses three different cloud providers for three different purposes, and justify the reasoning behind each choice.

Topic 6.3: Applications and Implications of Cloud Computing

1. Discuss in detail the major applications of cloud computing across different sectors.
 - a) Explain the role of cloud computing in data storage, web hosting and content delivery, and machine learning/AI, providing a real-world example for each.
 - b) Construct a scenario in which a small business relies on all three applications simultaneously, describing how each contributes to its operations.
 - c) Evaluate how cloud computing has changed the way businesses operate compared to relying solely on local, on-premises hardware.
2. Evaluate the key implications of cloud computing, focusing on security, cost, and compliance.
 - a) Discuss the main data security challenges associated with storing sensitive data on remote servers, and describe at least three security measures that can mitigate these risks.
 - b) Analyze how effective resource management and cost monitoring can help an organization avoid unnecessary cloud spending.
 - c) Discuss why compliance and regulatory issues can vary significantly by region and industry, and evaluate the challenges this creates for global organizations using cloud services.

Topic 6.4 & 6.4.1: Blockchain Fundamentals

1. Discuss in detail the fundamentals of blockchain technology, including its core principles and key components.
 - a) Explain decentralization, immutability, and consensus mechanisms, describing how each contributes to blockchain's overall security.
 - b) Construct a diagram or description showing how blocks are linked together using hashes and parent hashes, and explain what happens if someone attempts to alter data in an older block.
 - c) Evaluate why blockchain's peer-to-peer network structure makes it more resistant to a single point of failure compared to a traditional centralized database.
2. Analyze the role of Peer-to-Peer Networks in Blockchain in comprehensive detail.
 - a) Explain how peer-to-peer networks function, describing the dual role each node plays as both client and server.
 - b) Discuss why blockchain relies on peer-to-peer networking rather than a traditional client-server model.
 - c) Evaluate a scenario where a hacker gains control of one node in a 1,000-node blockchain network, and discuss what the hacker would actually need to control in order to successfully alter the network's official transaction history.

Topic 6.4.2 & 6.4.3: Use Cases, Cryptocurrencies, and Smart Contracts

1. Discuss the applications of blockchain in real-world scenarios, including cryptocurrencies, smart contracts, and product tracking.
 - a) Explain the role of cryptocurrencies in the digital economy, and describe how blockchain ensures transactions cannot be altered without needing a bank.
 - b) Describe how smart contracts work, using the vending machine analogy, and construct an example scenario (such as an escrow transaction) tracing each step of its automatic execution.
 - c) Evaluate the challenges associated with smart contracts, such as the need for error-free code, and discuss what could go wrong if a smart contract contained a coding bug.
2. Analyze the broader use cases of blockchain technology beyond cryptocurrency.
 - a) Describe how blockchain can be applied to supply chain management, healthcare, and voting systems, providing a specific benefit for each.
 - b) Evaluate which of these three use cases you believe would have the greatest positive impact on society, and justify your reasoning.
 - c) Discuss potential challenges or limitations that might arise when implementing blockchain in any one of these three areas.

Topic 6.5: Applications and Implications of Blockchain

1. Discuss in detail how blockchain enables transparent and secure tracking of products through a supply chain.
 - a) Explain how each transaction recorded on a decentralized ledger contributes to a product's traceability, from raw material supplier to final customer.
 - b) Construct a step-by-step example tracing a product (such as a cotton shirt) through multiple supply chain stages, describing what gets recorded on the blockchain at each step.
 - c) Evaluate why blockchain-based supply chain tracking might be more trustworthy to consumers than a simple printed label.
2. Analyze data security in blockchain using the "Sending a Secure Letter" analogy presented in this unit.
 - a) Explain how encryption, digital signatures, network verification, and decentralization each contribute to blockchain's overall data security, referencing the letter-sending analogy for each concept.
 - b) Discuss how blockchain is applied in financial services to make transactions faster and safer, particularly for international money transfers.
 - c) Evaluate which of the four security concepts (encryption, digital signature, network verification, decentralization) you believe would be hardest for a hacker to bypass, and justify your reasoning.

Topic 6.6: Future Trends and Innovations

1. Discuss the advancements and benefits of edge computing in modern technology.
 - a) Explain how edge computing differs from traditional centralized cloud computing, focusing on the concept of latency.
 - b) Construct a comparison scenario tracing the response time of an autonomous vehicle using traditional cloud processing versus edge computing, and discuss the real-world safety implications of the difference.
 - c) Evaluate at least two additional real-world applications (such as smart cities or healthcare monitoring) that would benefit significantly from edge computing's low latency.
2. Explain the concept of serverless architectures and their impact on application development.
 - a) Describe what "serverless" actually means, clarifying the common misconception that no physical servers are involved.
 - b) Construct a scenario in which a developer uses a serverless service (such as AWS Lambda) to build a small feature, describing how resources are automatically allocated and billed.
 - c) Evaluate the overall impact of serverless architectures on reducing operational complexity for developers, compared to traditional server management approaches.